

Movie Recommendation System

Project Report (Python)

1. Introduction

A recommendation system suggests items to a user based on their choices or on the properties of the items. Websites like Netflix and YouTube use big versions of this idea. In this project a small movie recommendation system was made in Python using only basic programming concepts.

2. Objective

The aim is to build a console program that can show movies, recommend them by genre or rating, list the top movies, and give a personal recommendation from the ratings the user enters. No advanced libraries or concepts are used.

3. Python Concepts Used

Concept	Where it is used
Lists	Store movie names, genres and ratings
for loop	Go through all movies and print matches
while loop	Main menu and input checking
if / elif / else	Menu choices and rating checks
Functions	Split the program into small parts
Bubble sort	Sort movies by rating for the top 5 list

4. Working of the Program

Step 1: The program stores 15 movies in three lists (name, genre, rating).

Step 2: A while loop shows the menu again and again until the user chooses Exit.

Step 3: An if-elif-else block checks the option and calls the matching function.

Genre recommendation: A for loop checks each movie and prints it if the genre matches.

Rating recommendation: The user enters a minimum rating and all movies with an equal or higher rating are printed.

Top 5: Copies of the lists are sorted with bubble sort, highest rating first.

Personal recommendation: The user rates seven movies from 1 to 5. Points are added to each genre. The genre with the most points is chosen and unrated movies of that genre are recommended.

5. Sample Output

```
Choose an option (1-6): 3
```

```
Enter minimum rating (0 to 10): 8.5

Movies with rating 8.5 or more:
* Inception ( 8.8 )
* The Dark Knight ( 9.0 )
* Interstellar ( 8.7 )
* Forrest Gump ( 8.8 )
* The Matrix ( 8.7 )
Total found: 5

Top 5 movies:
1 . The Dark Knight - 9.0
2 . Inception - 8.8
3 . Forrest Gump - 8.8
4 . Interstellar - 8.7
5 . The Matrix - 8.7

Your favourite genre looks like: Sci-Fi
Recommended for you (movies you have not rated):
* Interstellar ( 8.7 )
* The Matrix ( 8.7 )
```

6. Limitations

The movie list is small and fixed. The system does not learn from many users and it does not save data after the program closes. Only genre is used to find similar movies.

7. Future Scope

The project can be improved by reading movies from a file, saving user history, adding more details such as actor and year, and building a graphical interface.

8. Conclusion

The project shows that a working recommendation system can be built with only loops, conditions, lists and functions. It helps in understanding how real recommendation systems choose items for users.

Appendix: Source Code

```
# Simple Movie Recommendation System
# Uses only lists, loops, if/elif/else and while

# ----- Movie data (parallel lists) -----
names = ["Inception", "The Dark Knight", "Titanic", "The Notebook",
         "Avengers", "Interstellar", "Toy Story", "Finding Nemo",
         "The Conjuring", "Get Out", "Forrest Gump", "The Matrix",
         "Frozen", "Joker", "La La Land"]

genres = ["Sci-Fi", "Action", "Romance", "Romance",
         "Action", "Sci-Fi", "Animation", "Animation",
         "Horror", "Horror", "Drama", "Sci-Fi",
         "Animation", "Drama", "Romance"]

ratings = [8.8, 9.0, 7.9, 7.8,
           8.0, 8.7, 8.3, 8.2,
           7.5, 7.7, 8.8, 8.7,
           7.4, 8.4, 8.0]

genre_list = ["Sci-Fi", "Action", "Romance", "Animation", "Horror", "Drama"]

# ----- Helper functions -----
def show_all():
    print("\nAll movies:")
    print("-" * 45)
    i = 0
    while i < len(names):
        print(i + 1, ".", names[i], "|", genres[i], "|", ratings[i])
        i = i + 1

def by_genre():
    print("\nAvailable genres:")
    for g in genre_list:
        print("-", g)
    choice = input("Enter a genre: ")

    found = False
    print("\nMovies in", choice, ":")
    for i in range(len(names)):
        if genres[i].lower() == choice.lower():
            print(" ", names[i], "(", ratings[i], ")")
            found = True

    if found == False:
        print("Sorry, no movies found for that genre.")

def by_rating():
    limit = float(input("Enter minimum rating (0 to 10): "))
    if limit < 0 or limit > 10:
        print("Rating must be between 0 and 10.")
        return

    count = 0
    print("\nMovies with rating", limit, "or more:")
    for i in range(len(names)):
        if ratings[i] >= limit:
            print(" ", names[i], "(", ratings[i], ")")
            count = count + 1

    if count == 0:
        print("No movie matches this rating.")
    else:
        print("Total found:", count)

def top_movies():
    # copy the lists so the original order is not changed
    n = []
    r = []
```

```

for i in range(len(names)):
    n.append(names[i])
    r.append(ratings[i])

# bubble sort (highest rating first)
for i in range(len(r) - 1):
    for j in range(len(r) - 1 - i):
        if r[j] < r[j + 1]:
            temp = r[j]
            r[j] = r[j + 1]
            r[j + 1] = temp
            temp2 = n[j]
            n[j] = n[j + 1]
            n[j + 1] = temp2

print("\nTop 5 movies:")
for i in range(5):
    print(i + 1, ".", n[i], "-", r[i])

def personal_recommendation():
    print("\nRate these movies from 1 to 5 (enter 0 if you have not seen it).")
    picks = [0, 1, 2, 4, 6, 8, 10]      # movies we will ask about
    user_scores = []

    for p in picks:
        score = int(input(names[p] + " (" + genres[p] + "): "))
        while score < 0 or score > 5:
            print("Please enter a number from 0 to 5.")
            score = int(input(names[p] + " (" + genres[p] + "): "))
        user_scores.append(score)

    # add up the score for every genre
    genre_points = [0, 0, 0, 0, 0, 0]
    for k in range(len(picks)):
        g = genres[picks[k]]
        for x in range(len(genre_list)):
            if genre_list[x] == g:
                genre_points[x] = genre_points[x] + user_scores[k]

    # find the genre with the highest points
    best = 0
    for x in range(len(genre_points)):
        if genre_points[x] > genre_points[best]:
            best = x

    if genre_points[best] == 0:
        print("\nYou did not rate anything, so we cannot personalise yet.")
        return

    print("\nYour favourite genre looks like:", genre_list[best])
    print("Recommended for you (movies you have not rated):")

    shown = 0
    for i in range(len(names)):
        already_rated = False
        for k in range(len(picks)):
            if picks[k] == i and user_scores[k] > 0:
                already_rated = True

        if genres[i] == genre_list[best] and already_rated == False:
            print(" *", names[i], "(", ratings[i], ")")
            shown = shown + 1

    if shown == 0:
        print(" You have already seen everything in this genre!")

# ----- Main menu -----
print("=== Movie Recommendation System ===")
running = True

while running:
    print("\n1. Show all movies")
    print("2. Recommend by genre")
    print("3. Recommend by minimum rating")

```

```
print("4. Show top 5 movies")
print("5. Personal recommendation")
print("6. Exit")

option = input("Choose an option (1-6): ")

if option == "1":
    show_all()
elif option == "2":
    by_genre()
elif option == "3":
    by_rating()
elif option == "4":
    top_movies()
elif option == "5":
    personal_recommendation()
elif option == "6":
    print("Thank you for using the system. Goodbye!")
    running = False
else:
    print("Invalid choice, please try again.")
```